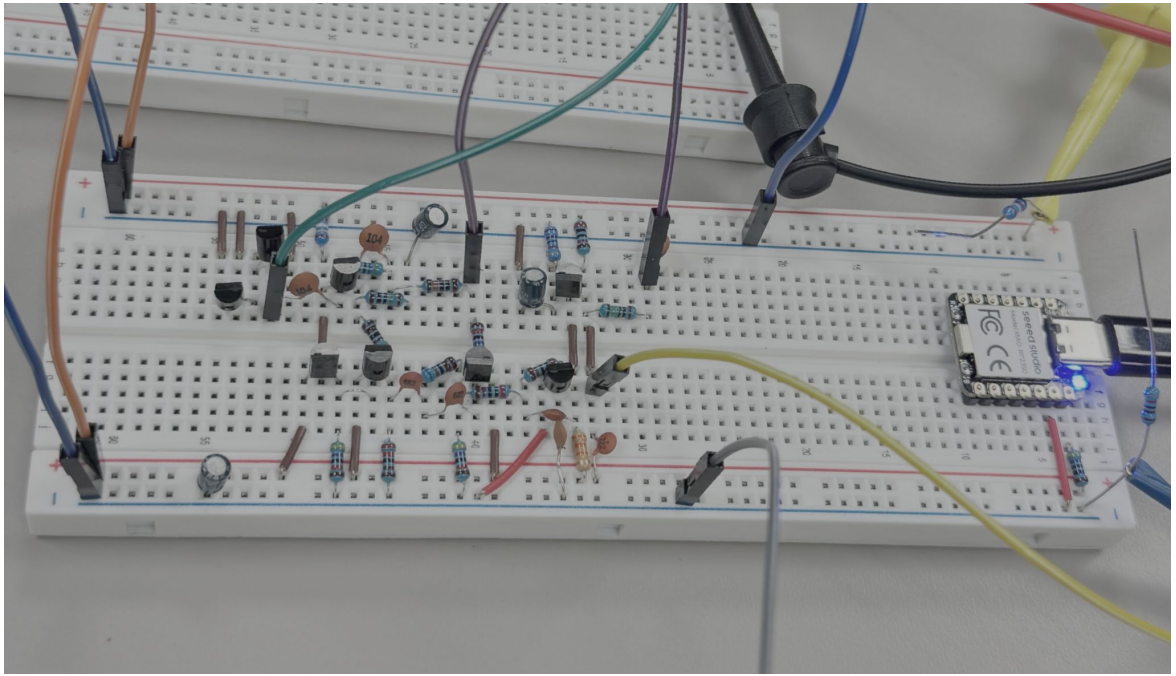
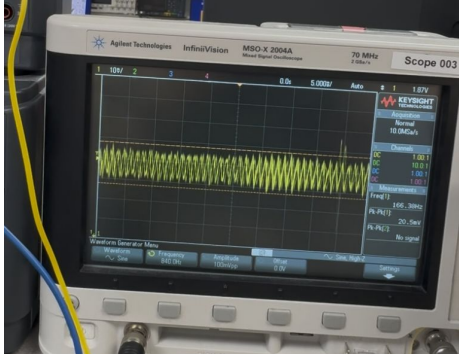
October 14th Progress Report

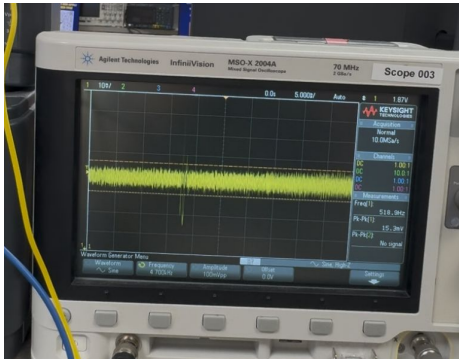
Ethan Ewer

Problem: Custom Amplifier has a lot of noise









Solution:

- Use an existing design that meets our requirements
- Add passive pre-amp and post-amp filters

Requirements:

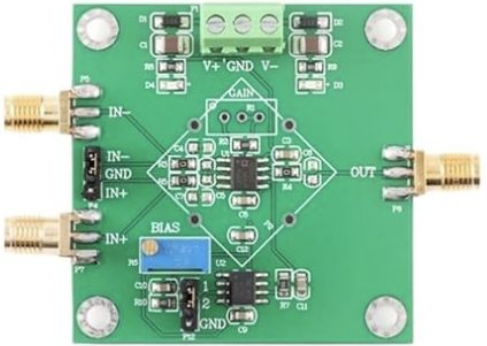
- Input noise under 4.5 nV/√Hz
- Input impedance over 1 MΩ
- 200 V/V gain
- Bandwidth flat up to 2.2 kHz

AD8421 selected in pre-built amplifier

Requirements:

- Input noise under 4.5 nV/√Hz (gets 3.2 nV/√Hz)
- Input impedance over 1 MΩ (gets 30 GΩ)
- 200 V/V gain (configurable from 1-10000 V/V)
- Bandwidth flat up to 2.2 kHz (yes)





Next Steps

- Wait for ordered amplifier to arrive (estimated arrival by 10/26) and then test it
- Test amplifier with ADC and MCU
- Test end-to-end device

Feedback

